

HELENA, MT, USA

WMO: 727720

Lat: 46.606N

Lon: 111.964W

Elev: 1167

StdP: 88.07

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24144

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-20.7	-27.9	0.3	-23.1	-24.6	0.5	-20.5	12.2	4.6	10.9	4.5	1.6	280	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	34.1	16.4	32.3	15.9	30.4	15.5	18.0	29.7	17.1	28.8	16.3	27.7	4.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	11.8	20.4	13.2	10.9	19.9	12.0	10.1	19.8	55.1	29.6	52.2	28.7	49.7	27.5	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	8.2	-28.3	37.0	4.7	1.4	-31.7	38.0	-34.4	38.8	-37.1	39.6	-40.5	40.6		
(5)			-28.3	19.9	4.2	1.3	-31.3	20.9	-33.7	21.7	-36.0	22.5	-39.0	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-4.8	-2.9	2.2	6.9	12.0	16.6	21.8	20.6	15.1	7.7	0.6	-4.7
	DBStd	10.72	7.59	7.23	6.19	4.43	4.31	4.00	3.25	3.21	4.51	4.77	6.33	6.98
	HDD10.0	2056	459	362	246	111	28	3	0	0	10	99	283	456
	HDD18.3	4148	717	595	501	342	201	78	8	17	114	331	531	714
	CDD10.0	1199	0	0	3	18	89	201	367	328	163	27	2	0
	CDD18.3	250	0	0	0	0	4	26	117	86	17	0	0	0
	CDH23.3	3566	0	0	0	10	108	406	1549	1164	317	12	0	0
	CDH26.7	1518	0	0	0	0	23	139	740	510	104	2	0	0
Wind	WSAvg	3.0	2.4	2.8	3.3	3.7	3.4	3.5	3.1	2.8	2.7	2.9	2.6	2.5
Precipitation	PrecAvg	289	13	9	16	26	45	51	30	32	26	16	12	12
	PrecMax	532	71	30	41	76	155	116	119	107	86	68	30	38
	PrecMin	159	0	0	0	2	7	2	2	0	2	0	1	0
	PrecStd	76	12	7	9	15	27	28	27	24	21	15	7	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	14.3	19.7	24.8	29.6	33.6	37.0	35.7	32.5	24.7	17.7	12.1
		MCWB	4.6	6.1	8.5	11.3	14.0	16.8	17.8	16.5	15.5	12.0	8.7	5.4
	2%	DB	8.3	10.5	16.3	21.3	26.4	30.5	34.9	33.5	29.6	21.3	13.8	8.6
		MCWB	3.3	4.5	7.1	9.8	13.2	16.2	16.8	16.0	14.3	10.7	6.6	3.6
	5%	DB	6.4	8.1	13.5	18.2	23.6	27.7	33.0	31.7	27.1	18.1	11.2	6.5
		MCWB	2.1	3.1	5.8	8.5	12.4	15.1	16.2	15.6	13.7	9.3	5.4	2.2
	10%	DB	4.6	5.9	10.9	15.2	20.9	25.4	31.1	29.7	24.4	15.5	9.0	4.5
		MCWB	1.0	1.9	4.6	7.2	11.2	14.1	15.8	15.2	12.8	8.2	4.2	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	6.8	9.4	12.3	15.3	18.6	19.6	18.7	17.1	13.0	9.6	5.8
		MCDB	10.0	13.3	18.6	22.5	25.9	29.0	32.3	30.4	26.8	22.8	16.6	11.3
	2%	WB	3.6	5.0	7.6	10.4	14.0	17.0	18.3	17.5	15.7	11.4	7.2	4.0
		MCDB	7.8	9.8	15.3	19.8	23.9	28.0	30.5	29.2	26.4	19.6	12.9	8.1
	5%	WB	2.3	3.4	6.1	9.0	12.9	15.8	17.5	16.6	14.6	10.0	5.8	2.5
		MCDB	6.0	7.7	12.8	17.3	22.1	25.8	29.6	28.3	25.0	17.4	10.8	6.1
	10%	WB	1.1	2.0	4.8	7.7	11.7	14.8	16.7	15.8	13.3	8.6	4.4	1.1
		MCDB	4.4	5.7	10.5	14.7	19.7	23.7	28.2	27.1	23.4	14.9	8.7	4.2
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.2	11.8	12.6	12.9	13.6	16.5	16.7	15.0	12.4	10.2	9.1
		MCDBR	11.0	12.8	16.1	18.2	18.0	18.1	19.2	20.0	19.9	17.3	13.5	10.5
		MCWBR	7.3	7.9	8.7	9.1	7.8	7.4	6.9	7.5	8.3	8.7	7.9	6.9
	5% WB	MCDBR	10.6	12.3	15.4	17.0	16.5	16.8	17.5	18.0	18.0	16.0	12.7	10.4
		MCWBR	7.3	8.0	8.6	8.7	7.4	7.4	6.9	7.1	7.8	8.3	7.7	7.1
Clear-Sky Solar Irradiance	taub	0.249	0.260	0.275	0.313	0.321	0.329	0.336	0.351	0.318	0.290	0.262	0.248	
	taud	2.489	2.492	2.554	2.455	2.493	2.503	2.497	2.425	2.497	2.572	2.565	2.509	
	Ebn at Noon	876	935	965	946	945	934	923	894	901	885	855	839	
	Edh at Noon	63	77	85	104	104	103	103	106	90	72	59	55	
All-Sky Solar Radiation	RadAvg	1.51	2.49	3.69	4.79	5.57	6.46	7.16	6.03	4.42	2.76	1.61	1.17	
	RadStd	0.13	0.18	0.25	0.29	0.49	0.47	0.40	0.33	0.39	0.23	0.15	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.37	-0.67	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.57	N/A	N/A	N/A	N/A

Nomenclature: See separate page